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Effects of Acute Thermal Stress on the Survival, Predator Avoidance, and Physiology of Juvenile Fall Chinook Salmon

Abstract

We subjected juvenile fall chinook salmon from the Hanford Reach of the Columbia River to acute thermal stressors in the laboratory that were derived from field data. We assessed the effects of thermal stress on: (1) the extent of direct mortality; (2) the vulnerability of fish to predation by smallmouth bass; and (3) some general physiological stress responses and synthesis of heat shock protein 70 (hsp70). Thermally-stressed fish showed little direct mortality and no increases in vulnerability to predation. However, these fish showed transient increases in plasma concentrations of cortisol, glucose, and lactate, and a dramatic (25-fold higher than controls) and persistent (lasting 2 wk) increase in levels of liver hsp70. Our results indicate that exposure of Hanford Reach juvenile fall chinook salmon to such stressors did not lead to significant increases in direct mortality or vulnerability to predation, but did alter physiological homeostasis, which should be of concern to those managing this resource. Because our fish received only a single exposure to one of the stressors we examined, we are also concerned about the consequences of exposing fish to multiple, cumulative stressors—a likely scenario for fish in the wild.

Introduction

The Hanford Reach, the last free-flowing section of the Columbia River, supports a large and important population of wild fall chinook salmon (*Oncorhynchus tshawytscha*). Recently, concerns have been raised about potential losses of these fish due to river level fluctuations resulting from power peaking operations at dams upstream of the reach. Rapid fluctuations in river level can cause stranding (the trapping of fish on or within the unwatered substrate as a result of receding river level) of juvenile fall chinook salmon and lead to direct mortality. For example, during spring and early summer weekly sampling of fall chinook salmon on the Hanford Reach in 1998, almost 8,000 fish were found stranded and dead along the shoreline (Nugent et al. 2001a).

Besides stranding, fish can also become entrapped in isolated backwaters that form when water rapidly recedes. In the spring and summer of 1998, the Washington Department of Fish and Wildlife (WDFW) estimated that at least 24,000

juvenile fall chinook salmon were entrapped for various lengths of time in isolated pools (Nugent et al. 2001a). In 1999, using random plot sampling and computer modeling, they estimated that at least 125,000 fall chinook salmon perished due to stranding and entrapment (Nugent et al. 2001b). Although such entrapment may not cause rapid mortality, it may lead to delayed mortality from two indirect sources: predation or alterations in performance caused by sublethal heat stress. We have observed predation by birds and fish on juvenile salmon entrapped in pools on several occasions in the Hanford Reach.

Alterations in performance of fish caused by sublethal heat stress is a more insidious and potentially devastating effect of entrapment. Fish entrapped in backwaters on the Hanford Reach can experience a unique type of thermal stressor, typically consisting of exposure to gradual (i.e., over several hours) and extreme (i.e., >10°C above ambient temperatures) increases in temperature followed by an abrupt decrease in temperature when they are liberated into the main river as water levels rapidly rise. Sublethal, acute thermal shocks increase the vulnerability of fish to predation (Sylvester 1972, Coutant 1973, Yocom and Edsall 1974, Deacutis 1978), cause a variety of physiological disturbances in fish including increases

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in plasma glucose and cortisol (Wedemeyer 1973, Strange et al. 1977, Crawshaw 1977, 1979, Bailey et al. 1991, Soncini and Glass 1997), and elicit the rapid synthesis of stress proteins that play a role in repair and protection from environmentally induced cellular damage (Sanders 1993, Iwama et al. 1998). However, information from these studies may not apply to fall chinook salmon in the Hanford Reach because of their species, age, and the unique nature of the stressor experienced by these fish. Clearly, such a stressor has the potential to alter behavioral, physiological and cellular mechanisms that could lead to delayed or indirect mortality of the fish. An understanding of the effects of acute heat stress on juvenile fall chinook salmon is necessary to assess the impacts of entrapment on populations of these fish in the Hanford Reach, and it will also contribute to knowledge of thermal stressors in general.

In this study, we exposed juvenile fall chinook salmon from the Hanford Reach to four heat stress scenarios in the laboratory that were derived from field data. We then assessed the effects of thermal stress on: (1) the extent of direct mortality; (2) the vulnerability of fish to predation by smallmouth bass (*Micropterus dolomieu*); and (3) some general physiological stress responses and synthesis of heat shock protein 70 (hsp70).

Methods

Test fish

Subyearling fall chinook salmon were collected by seine from the Hanford Reach during April and May in 1998 and 1999. Fish were transported to our laboratory in a truck equipped with a large, insulated aluminum tank and aerated water. They initially averaged 38.4 ± 0.6 mm in length and 0.7 ± 0.1 g in weight. The fish were placed in 1400 L flow-through circular tanks, reared outdoors under ambient photoperiod, and fed twice daily (at a ration of 3-4 % body weight per day) with blood worms for the first month and a commercial starter feed thereafter.

Sixteen smallmouth bass (mean length = 306 ± 6 mm; weight = 457 ± 27 g) were used in our predation trials and were collected by angling from the lower Yakima River, Washington, in 1998 and by electrofishing in the Columbia River in 1999. They were transported to our laboratory as previously described and placed four fish per tank in each of two 3.75-m-diameter, 1-m-deep circular

tanks. The tanks were lined with gravel and cobble substrates and included four pieces of 0.25-m-long, 15-cm-diameter PVC pipe randomly scattered along the bottom to serve as cover. These tanks were indoors and served as our predation arenas. They were surrounded with curtains to minimize outside disturbance and under an ambient photoperiod simulated with 25 W incandescent bulbs and timers to produce a gradual intensity dawn and dusk. Predators were fed a maintenance diet of juvenile chinook salmon (about two fish per predator every 2-3 d). All fish were acclimated to and held in well water heated to 12°C, which was representative of river temperatures in the Hanford Reach during spring (Nugent et al. 2001a, b). The water was heated using large, single-pass electric heaters and excess dissolved gas generated by heating was dissipated by a packed column.

Direct mortality bioassays

We used fall chinook salmon in these tests, which were conducted during June and July in 1999, that had a mean length of 55.5 ± 0.4 mm and a weight of 1.5 ± 0.1 g ($n = 160$). Four groups of 20 fish were serially removed from their holding tank and transferred to one of four 0.61-m-diameter circular test tanks. Two of these tanks contained a 2000 W immersion heater connected to an electronic controller that was used to produce the thermal stressors (described below). Our thermal stressors were modeled after data from thermographs placed in entrapment pools in the Hanford Reach (see examples in Figure 1A; Nugent et al. 2001a, b). All tanks received the same water as the holding tanks with an inflow of about 1 L/min and a volume of 104 L. Densities in the test tanks were ~ 0.28 g/L. Fish were held undisturbed for at least 2 d prior to starting the bioassays.

At the start of a bioassay, fish in the two heated tanks were designated as treatment fish, whereas fish in the other two tanks remained at 12°C during the experiment and were designated as controls. At 0700, treatment fish were subjected to one of four thermal stressors in separate trials (Figure 1B). Stressor 1 consisted of heating the water from 12°C to 26°C at a rate of 2°C/hr, followed by a decrease to 12°C in about 1 hr. Stressor 2 was the same, except that the temperature increase was 4°C/hr. For these stressors, fish remained at 26°C for about 10-15 min. Stressor 3

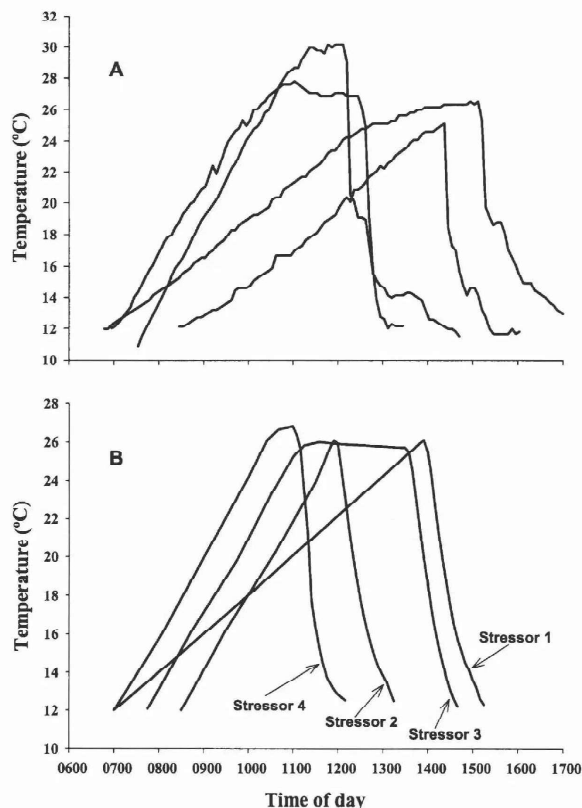


Figure 1. (A) Temperature profiles in selected entrapment pools on the Hanford Reach during spring 1998. Data were taken from thermographs placed in the pools. (B) Thermal stressors used in laboratory experiments. Curves were spline-fit to thermograph data from several tanks during our study. Starting times for the curves were staggered to facilitate graphical representation.

was the same as Stressor 2, except that fish remained at 26°C until about 25% of the fish showed disequilibrium or died. Stressor 4 was also similar to Stressor 2, except that the maximum temperature was increased to approximately 27°C. For all stressors, when heating was complete, the heaters were turned off and a 6–7°C inflow of water was used to cool water temperature to 12°C in about 1 hr. After the treatments, fish remained in their tanks at 12°C for a period of 5 d. Miniature temperature loggers in each tank recorded water temperature every 5 min during application of the stressor and every hour thereafter. At each hour during the bioassay and three times a day thereafter, we removed and tallied dead fish. In total, we conducted one complete trial for each stressor consisting of two replicate tanks each of treatment and control fish.

Predation Trials

We conducted predation trials during two years, from July through September in 1998, and from June through July in 1999. In 1998, fall chinook salmon had a mean length of 77.0 ± 0.4 mm and a weight of 4.7 ± 0.1 g ($n = 279$); in 1999, fish had a mean length of 56.0 ± 0.2 mm and a weight of 1.6 ± 0.1 g ($n = 170$). For a trial, four groups of 15–20 fish were anesthetized using 50 mg/L tricaine methanesulfonate (MS-222) buffered with sodium bicarbonate. Two groups were designated as control and two were designated as treatment. We randomly selected and marked two groups (i.e., either control or treatment fish) by clipping the adipose fin with surgical scissors, whereas the other groups were sham marked by simply holding them out of the water for a sufficient period. Fourteen marked fish were placed in each of two control and treatment tanks as described in the mortality bioassays. Fish were held in these tanks for 2–3 d before being subjected to thermal stress.

On the morning of a predation trial, fish in the treatment tanks were subjected to one of two thermal stressors. Control fish and the predators always remained at 12°C. In 1998, we conducted eight trials using Stressor 1 and 12 trials using Stressor 2. In 1999, we conducted 12 trials using Stressor 1. Immediately after fish were exposed to thermal stress (i.e., when water temperature in the treatment tanks was back at 12°C), water level in all tanks was dropped to approximately 16 L. Fish were then released into a 19 L bucket below the tanks. Equal numbers of control and treatment fish were mixed in one bucket and poured into a predation tank. Smallmouth bass were allowed to consume juvenile salmon until 50% of the prey were eaten or 2 hr had passed, whichever occurred first. We observed fish behavior and estimated the number of prey eaten. Afterwards, the remaining prey were seined from the tanks and placed in a lethal dose (200 mg/L) of MS-222. Lengths and weights were recorded and fish were identified as treatment or control based on the presence of an adipose fin.

We analyzed these data in a manner identical to that of Mesa (1994). We first subjected data to a heterogeneity χ^2 analysis to determine if the individual tests were homogenous (Sokal and Rohlf 1981). Chi-squared goodness-of-fit tests were then used on pooled data to determine if predation was random (i.e., 50:50) on treated versus control fish.

Because the χ^2 test has low statistical power (i.e., low probability of correctly rejecting the null hypothesis) when relatively small sample sizes are used, we set the significance level at $P = 0.10$, the detectable difference in predation rates (i.e., the effect size) at 20%, and target sample sizes at about 180 to increase power and reduce the probability of the more serious type II error (Fairbairn and Roff 1980, Peterman 1990). Tests where less than 30% or more than 70% of the prey were eaten were excluded from the analysis for reasons discussed in Mesa and Warren (1997).

Physiology Experiment

To assess some physiological responses of juvenile fall chinook salmon to acute thermal stress, we conducted one experiment in September 1999. Fish used in this experiment had a mean length of 90.5 ± 0.5 mm and a mean weight of 7.7 ± 0.1 g ($n = 126$). Forty-five fish were stocked in each of two treatment tanks and two control tanks, as previously described. Fish were held in these tanks for 7 d prior to applying the stressor. At 0730 on the eighth day, fish in treatment tanks were subjected to Stressor 1; control fish remained at 12°C during the experiment. Just prior to the stressor, we removed four fish from each tank (time = 0) and also removed four fish at the following times after the start: 3 hr (midway point of the temperature increase); 7 hr (maximum temperature); 9 hr (1 hr after the rapid temperature decrease); and at 24, 72, 168, and 336 hr. All sampled fish were placed in a lethal dose of buffered MS-222, weighed and measured, and blood samples collected into ammonium-heparinized capillary tubes after the caudal peduncle was severed. Plasma was obtained by centrifugation and stored at -80°C for future analysis. Plasma cortisol was measured using an enzyme-linked immunosorbent assay (ELISA) modified from procedures described by Munro and Stabenfeldt (1984). Plasma lactate and glucose were assayed using commercial kits (Sigma Diagnostics) modified by us for use with microplates. Due to an insufficient amount of plasma, we were unable to assay all analytes from some fish. After collecting blood, we excised the liver, weighed it, wrapped it in aluminum foil and placed it in liquid nitrogen. These samples were stored at -80°C for future analysis of hsp70 expression using an ELISA (Forsyth et al. 1997).

We calculated the mean and SE for all plasma constituents at each sample period and plotted them

over time. Within each sample period, we compared the means of treatment and control fish using two-sample t -tests for use with unequal variance. We expressed the hsp70 content of liver samples in relative units as a percentage of a positive control and analyzed the means as just described. For these tests, the level of significance was set at $P = 0.05$.

Results

Direct Mortality Bioassays

Only one control fish died during the trial exposing fish to Stressor 1, whereas only one treatment fish died during the trial assessing Stressor 2. For fish exposed to Stressor 3, about 2–2.5 hr after the maximum temperature had been reached were required for about 25% of the fish to die. Thereafter, only one treatment fish died on the second day. For fish exposed to Stressor 4, eight fish died in one treatment tank and five fish died in the other, all before the cool down phase of the stressor was complete. Thereafter, only one more treatment and one control fish died.

Predation Trials

In 1998, we conducted a total of eight trials using fish exposed to Stressor 1 with an average of 48% of fish eaten per trial. A total of 53 control and 54 treatment fish were eaten, which did not differ significantly from random (Table 1). Statistical power (to detect a 20% difference in predation rates) of the pooled χ^2 test was 0.66. Duration of predation trials ranged from 5 min to 2 hr, however the number of control and treatment fish consumed was independent of trial duration (χ^2 test of independence). There were no significant differences in length or weight of control and treatment fish measured after the trials.

Also in 1998, we conducted a total of 12 trials using fish exposed to Stressor 2 with an average of 47.5% of fish eaten per trial. There were a total of 67 control and 82 treatment fish eaten, which again was not significantly different from random (Table 2). Statistical power of the pooled χ^2 test was 0.79. Duration of predation trials ranged from 10 min to 2 hr, and again the number of control and treatment fish consumed was independent of trial duration. There was one trial with a minor (t test, $P < 0.05$) difference in length of control and treatment fish.

TABLE 1. Predation by smallmouth bass on juvenile fall chinook salmon acclimated to 12°C, subjected to a heating of 2°C/hr up to 26°C, and a temperature decrease within 1 hr to 12°C (i.e., Stressor 1 in Figure 1B). Data are from experiments conducted in 1998. No predation rates differed significantly from random (50:50, treatment:control).

Replicate or statistic	Number eaten		Percent eaten	Statistics	
	Control	Treatment		df	χ^2
1	7	5	43	1	0.333
2	4	6	36	1	0.400
3	11	8	68	1	0.474
4	4	7	46	1	0.818
5	10	5	54	1	1.667
6	6	8	50	1	0.286
7	5	8	46	1	0.692
8	6	7	46	1	0.077
Total	53	54		8	4.747
Pooled				1	0.009
Heterogeneity				7	4.738

TABLE 2. Predation by smallmouth bass on juvenile fall chinook salmon acclimated to 12°C, subjected to a heating of 4°C/hr up to 26°C, and a temperature decrease within 1 hr to 12°C (i.e., Stressor 2 in Figure 1B). Data are from experiments conducted in 1998. No predation rates differed significantly from random (50:50, treatment: control).

Replicate or statistic	Number eaten		Percent eaten	Statistics	
	Control	Treatment		df	χ^2
1	5	4	56	1	0.111
2	5	9	50	1	1.143
3	8	6	52	1	0.286
4	8	7	53	1	0.067
5	4	5	32	1	0.111
6	5	8	46	1	0.692
7	7	10	65	1	0.529
8	5	9	50	1	1.143
9	7	4	39	1	0.818
10	6	8	58	1	0.286
11	3	6	32	1	1.000
12	4	6	38	1	0.400
Total	67	82		12	6.586
Pooled				1	1.510
Heterogeneity				11	5.076

In 1999, we conducted a total of 12 trials using fish exposed to Stressor 1 with an average of 48.5% of fish eaten per trial. There were a total of 78 control and 83 treatment fish eaten, which was not significantly different from random (Table

TABLE 3. Predation by smallmouth bass on juvenile fall chinook salmon acclimated to 12°C, subjected to a heating of 2°C/hr up to 26°C, and temperature decrease within 1 hr to 12°C (i.e., Stressor 1 in Figure 1B). Data are from experiments conducted in 1999. No predation rates differed significantly from random (50:50, treatment:control).

Replicate or statistic	Number eaten		Percent eaten	Statistics	
	Control	Treatment		df	χ^2
1	5	6	39	1	0.091
2	2	7	32	1	2.778
3	12	8	71	1	0.800
4	7	8	54	1	0.067
5	4	8	50	1	1.333
6	4	6	36	1	0.400
7	7	8	54	1	0.067
8	6	5	39	1	0.091
9	8	8	57	1	0.000
10	9	5	50	1	1.143
11	6	8	50	1	0.286
12	8	6	50	1	0.286
Total	78	83		12	7.341
Pooled				1	0.155
Heterogeneity				11	7.186

3). Statistical power of the pooled χ^2 test was 0.81. Duration of predation trials ranged from 14 min to 2 hr, and again the number of control and treatment fish consumed was independent of trial duration. Although there were no significant differences in size of control and treatment fish used in these trials, fish used in 1999 were significantly smaller than those used in 1998 (*t*-tests, $P < 0.05$).

Physiology Experiment

Plasma cortisol levels in treatment fish were significantly higher than control fish prior to application of Stressor 1 (for unknown reasons) and also at 7 hr after the start of the stressor, which was the time of maximum temperature (Figure 2A). Thereafter, cortisol levels did not differ between treatment and control fish, except for a slight elevation in treatment fish at 72 hr. Plasma glucose levels increased gradually in treatment fish, peaking and becoming significantly higher than control fish only during the time of maximum temperature (Figure 2B). Control fish did show a dramatic and inexplicable increase in glucose levels at 9 hr. Lactate levels in treatment fish showed about a threefold increase during application of the stressor and were significantly higher than control fish at 7 and 9 hr (Figure 2C). Plasma

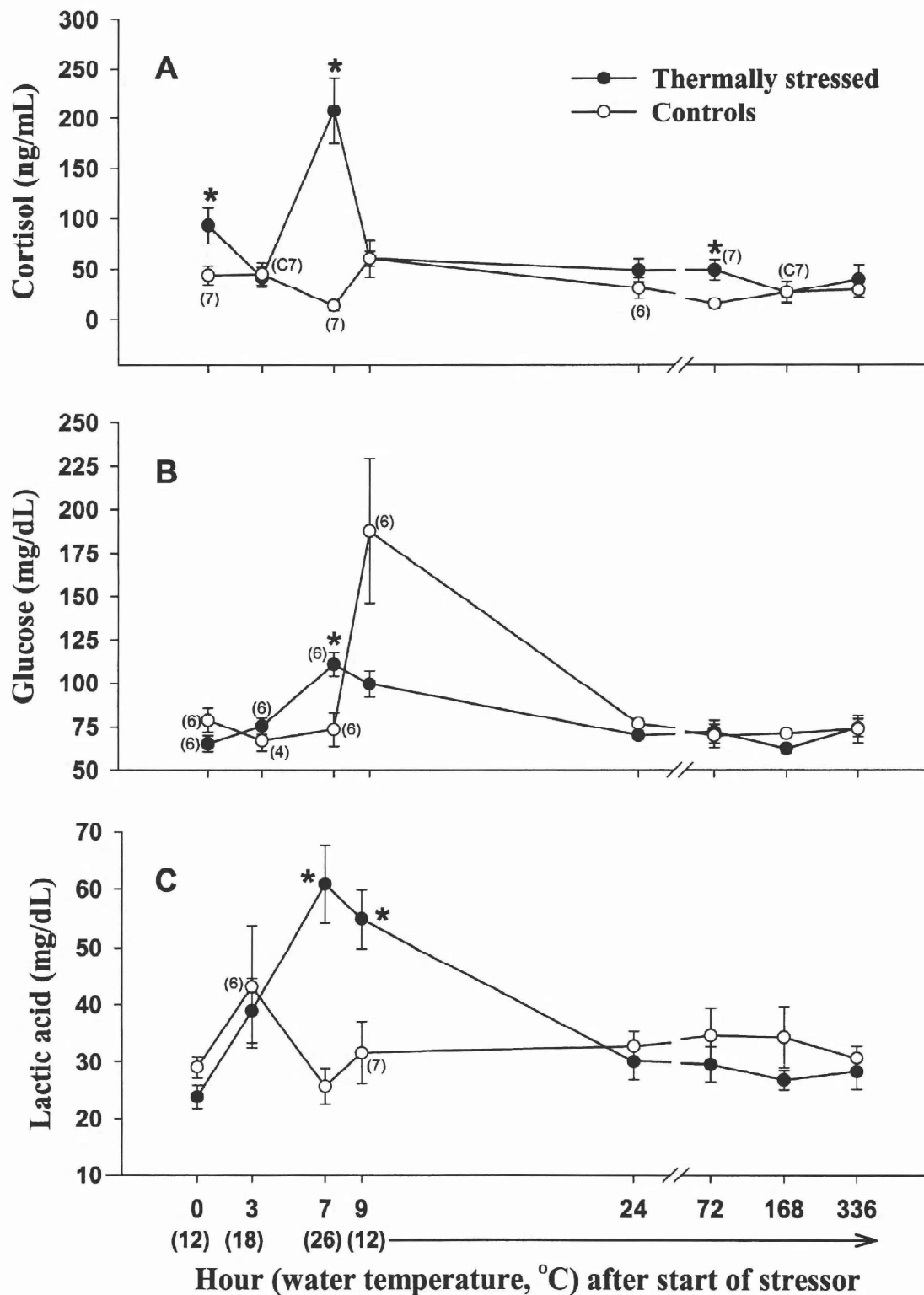


Figure 2. Mean ($\pm$ SE) plasma (A) cortisol, (B) glucose, and (C) lactate concentrations for fish at selected time intervals after they had been exposed to Stressor 1 and for unstressed controls. Means were derived from eight fish unless otherwise noted parenthetically; C = control fish. An asterisk denotes a significant difference between treatment and control fish at that time interval.

lactate concentrations did not differ between treatment and control fish at any sample period from 24 hr on. Liver hsp70 content displayed a dramatic increase during the experiment that persisted for the entire two week period (Figure 3). Levels of hsp70 in treatment fish were significantly higher than control fish at all sample periods except the first two.

Discussion

Our results indicate that juvenile fall chinook salmon exposed to naturally-occurring thermal stress showed some physiological changes (Figure 2), a dramatic increase in levels of liver hsp70 (Figure 3), but no increase in vulnerability to predation and little direct mortality (Tables 1-3). These thermal stressors, which were derived from field data and included a 14°C temperature increase over several hours followed by a rapid decrease

to ambient temperature, are unique stressors that, to our knowledge, have not been previously studied. Such thermal stress may be common in developed and highly managed river systems. Most research on thermal stress in fishes has focused on two broad areas: (1) examining the effects of acute temperature shocks, where fish acclimated to a certain temperature are subjected to dramatic changes in temperature usually within a few minutes (Coutant 1973, Wedemeyer 1973); and (2) examining the effects of different acclimation temperatures on selected measures of fish performance (Glova and McInerney 1977, Fuiman 1991, Cech et al. 1994). The wide variety of results produced by such studies indicates that they may not be useful for predicting the effects of acute thermal stressors, such as the types we evaluated, on fish. Indeed, the influence of thermal stressors on fish performance seems dependent on

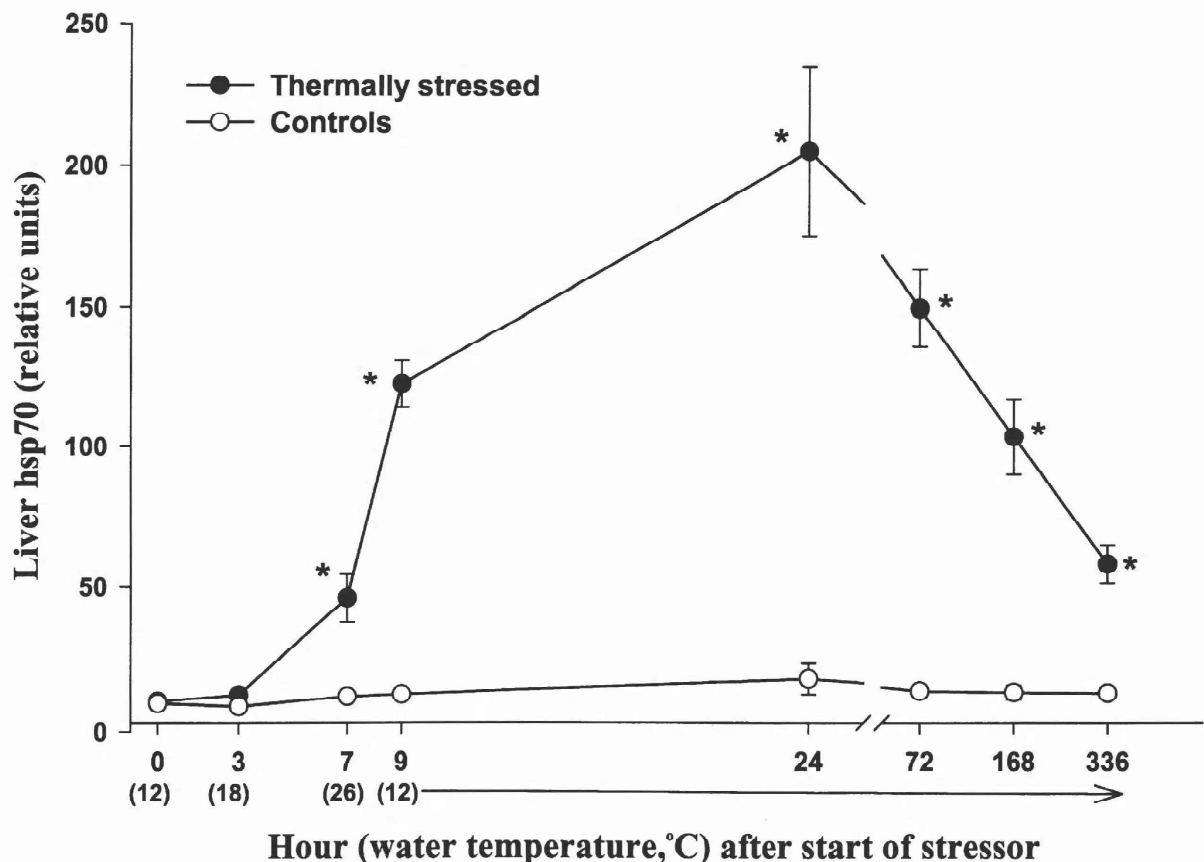


Figure 3. Mean ($\pm$ SE; $n = 8$) liver hsp70 content for fish at selected time intervals after they had been exposed to Stressor 1 and for unstressed controls. An asterisk denotes a significant difference between treatment and control fish at that time interval.

numerous factors, including species, size and age, biology and physiology of the organism (e.g., eury- vs. stenothermal species), acclimation temperature, magnitude of temperature change, and tempering rate, among others.

Few juvenile fall chinook salmon died after exposure to our acute thermal stressors up to 26°C (i.e., Stressors 1 and 2), a temperature within the lethal range for this species (Brett 1952, Templeton and Coutant 1971). We attribute the lack of significant mortality in fish to the short time fish remained at 26°C and use of a slow tempering rate. Tempering rate influences the critical thermal maximum and survival of fish (Elliott and Elliott 1995, Clapp et al. 1997), and slow tempering may prolong the time of death (Mather and Wahl 1989). Although our tempering rates were much slower than most reported in the literature, they represent what these fish would experience in the wild and probably helped prolong the time of death. However, our results indicate that if fish remain at 26°C (i.e., Stressor 3) or are exposed to temperatures slightly above 26°C (i.e., Stressor 4), mortality quickly ensues. Thus, for fish in the wild, the presence and use of cooler water refugia may be critical to their survival. Our conclusion agrees with Mundahl (1990), who reported that survival of 14 fish species in shrinking stream pools depended on the use of cooler shaded refugia. Our results suggest that survival of fish trapped in pools may be high if they can tolerate periods of high temperatures until the pool re-floods with cool water.

Besides death, loss of equilibrium is a common response of fish to a sublethal exposure to lethal temperatures. In fact, only slight excursions of temperature above or below those normally encountered can elicit a sequence of behavioral and locomotory symptoms of sublethal thermal injury (Logue et al. 1995). Coutant (1969) argued that equilibrium loss is important for fish subjected to thermal stress because loss of equilibrium occurs sooner than direct mortality and may make fish particularly vulnerable to predation. Indeed, loss of equilibrium and aberrant behavior are often mentioned as probable causes of the increased vulnerability to predation of fish subjected to acute thermal shocks (Sylvester 1972, Coutant 1973). However, the acute thermal stress our fish received (i.e., Stressors 1 and 2) did not sufficiently compromise behavioral and performance factors that affect the vulnerability of prey

fish to predation (Tables 1-3; Bams 1967, Coutant 1973, Mesa et al. 1994). Although we noted some aberrant behavior in fish as they approached the maximum temperature (e.g., erratic swimming or agitation), we saw no apparent behavioral differences in our treatment and control fish upon release to predators. Also, size of prey did not influence the outcome of our predation experiments. All fish responded to predation pressure by forming schools soon after attacks started. Thus, aside from species, age, or size differences, there appear to be differences in the nature and severity of acute thermal shocks and the type of thermal stress we exposed our fish to, at least with regard to predator avoidance ability. Juvenile fall chinook salmon were capable of avoiding predators soon after experiencing a thermal stressor of the type we used (Tables 1-3), a conclusion also noted in studies subjecting juvenile salmonids to physical stress (Olla and Davis 1989, Mesa 1994).

The physiological factors we examined indicated that exposure of our fall chinook salmon to an acute temperature increase was stressful. Plasma cortisol, glucose, and lactate all increased during the thermal warming, reached maximal values at the time of maximal temperature, and returned to levels similar to initial values by 24 hr (Figure 2). Some studies have reported similar responses, in terms of hypercortisolemia or hyperglycemia (or both), in other species exposed to thermal stress (Wedemeyer 1973, Strange et al. 1977, Kindle and Whitmore 1986, Thomas et al. 1986, Davis and Parker 1990). In contrast, some studies have shown little or no response of plasma cortisol and glucose in fish to thermal stress (Strange et al. 1977, Carmichael et al. 1984, Thomas et al. 1986). Thus, the physiological responses of fish to thermal stressors vary widely and may depend on species and type of stressor used. If plasma cortisol and glucose increase in response to thermal stress, they probably serve to mobilize energy stores for essential reactions (e.g., ATP production) and help maintain osmoregulatory function, among other things.

Plasma lactate concentrations were also higher in treatment fish relative to controls, significantly so at the time of maximum temperature and 1 hr after the cool down had occurred (Figure 2). We are unaware of other studies reporting changes in plasma lactate concentrations in fish subjected to thermal stress. Such elevations in plasma lactate indicate some degree of anaerobic metabo-

lism was occurring in fish exposed to the thermal stressor and are probably due to a combination of factors. During temperature increases in ectotherms, blood pH decreases and PCO_2 increases (Crawshaw 1979). At high temperatures and low pH, hemoglobin has a low affinity for oxygen (Schmidt-Nielsen 1991) and arterial PO_2 may decrease (Soncini and Glass 1997), thus leading to problems with oxygen delivery to the tissues. This, coupled with the increased metabolic demands and hyperactivity of our fish and the low oxygen carrying capacity of water at high temperatures, probably contributed to their lactacidosis. Such changes in the acid-base balance of fish subjected to temperature change probably contributes to mortality as lethal temperatures are approached.

Exposure of juvenile fall chinook salmon to Stressor 1 elicited a dramatic response in liver hsp70 content (Figure 3). The magnitude of maximum change represented a more than 25-fold increase over control levels. Mazur (1996) also reported that hsp70 levels in the gills of cutthroat trout were still elevated up to 3 wk after fish were exposed to a 2 hr, 15°C heat shock (7.4–22.4°C). Heat shock proteins (hsps) are a group of proteins synthesized by cells in response to heat and many other types of stressors, such as exposure to contaminants (Lindquist 1986, Lindquist and Craig 1988, Iwama et al. 1998). The constitutive and inducible expression of hsps is perhaps the most highly conserved genetic system known, existing in all organisms from bacteria to plants

and animals. The functions of hsps in response to stress help to maintain cellular homeostasis and include protein assembly, correct folding and translocation, and regulating hormone-receptor interactions (Iwama et al. 1998). The hsps prevent the aggregation and facilitate the renaturation of heat-damaged proteins following a heat shock, thus they play a vital role in thermotolerance (Tomanek and Somero 1999). In short, the hsp70 response of our fish probably provided cellular protection and enhanced survival against a thermal insult that approached the upper lethal temperature limits for this species.

In summary, we have shown that exposure of juvenile fall chinook salmon to naturally occurring thermal stressors elicited some acute physiological stress responses and a significant and prolonged elevation of liver hsp70 content, but did not lead to substantial mortality or increased vulnerability to predation. The physiological changes seen in our fish, and the consequences of exposing fish to multiple stressors, should be of concern to those charged with management of the Hanford Reach area.

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